

STUDY PROTOCOL

Open Access



The effect of a sustainable diet and time-restricted eating and probiotic in heart failure with reduced ejection fraction patients: study protocol for a randomized, double-blind, and placebo-controlled trial

Pegah Rahbarinejad^{1,2}, Sara Amini³, Asma Afshari¹, Seyyed Reza Sobhani⁴, Kayvan Sadri⁵, Golnaz Ranjbar¹, Majid Khadem Rezaian⁶, Farveh Vakilian³, Mohammad Hadi Eskandari⁷ and Reza Rezvani^{1,8*}

Abstract

Background Heart failure is a complex clinical syndrome characterized by structural and/or functional abnormalities that impair cardiac output, leading to alterations in cardiac biomarkers, metabolic disturbances, and gut dysbiosis. Lifestyle interventions, particularly dietary strategies such as time-restricted eating (TRE) and healthy diets like the sustainable diet (SD), along with probiotic supplementation, have been proposed as key components in the management of heart failure. Thus, the present study aims to evaluate the effects of co-administration of SD and TRE, either alone or in combination with *Lactobacillus plantarum* 299v supplementation, compared to control group, on cardiovascular function in overweight or obese patients with heart failure with reduced ejection fraction.

Methods We will conduct a randomized, double-blind, placebo-controlled, three-arm clinical trial over 60 days. Participants will be randomly assigned in a 1:1:1 ratio to one of three groups for 60 days: (A) co-administration of SD and TRE with probiotic supplementation ($n = 25$); (B) co-administration of SD and TRE with placebo supplementation ($n = 25$); and (C) control group with placebo supplementation ($n = 25$), serving as the control group. The primary outcomes will be changes in N-terminal pro-B-type natriuretic peptide and left ventricular ejection fraction. Secondary outcomes will include changes in fasting blood glucose, lipid profile, liver and kidney function markers, anthropometric indices, arterial stiffness, electrocardiogram indices, echocardiographic indices, sleep quality, appetite, energy intake, quality of life, exercise tolerance, and Zonulin and Trimethylamine N-oxide as gut-related biomarkers. All outcomes of this study will be evaluated before and after the intervention.

Discussion We hypothesize that the co-administration of SD and 10-hour TRE, either alone or in combination with probiotic supplementation, will result in weight loss and improvements in metabolic, hormonal, and

*Correspondence:

Reza Rezvani

Reza.Rezvani.rr@gmail.com; RezvaniR@mums.ac.ir

Full list of author information is available at the end of the article



© The Author(s) 2025. **Open Access** This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by-nc-nd/4.0/>.

echocardiographic markers in overweight and obese patients with heart failure with reduced ejection fraction compared to control individuals.

Ethical aspects The present trial received approval from the Medical Ethics Committee of Mashhad University of Medical Sciences, Mashhad, Iran (IR.MUMS.REC.1402.259; approved on December 16, 2023).

Trial registration Iranian Registry of Clinical Trials (IRCT), IRCT20231225060520N1. Registered 23 January 2024, <https://irct.behdasht.gov.ir/trial/74743>.

Keywords Sustainable diet, Time-restricted eating, Probiotics, Heart failure

Introduction

Heart Failure (HF)

HF is a complex, life-threatening clinical syndrome characterized by structural and/or functional cardiac abnormalities (inability to fill and/or pump blood) that impair cardiac output, resulting in inadequate circulation at normal physiological pressure [1]. The global prevalence of HF is estimated to be 64.3 million people [2]. According to the most recent guidelines [2, 3], HF is classified based on left ventricular ejection fraction (LVEF) into three phenotypes: reduced (HFrEF, $\leq 40\%$), mildly reduced (HFmrEF, 41–49%), and preserved (HFpEF, $\geq 50\%$), with HFrEF accounts for approximately half of all HF cases. N-terminal pro-B-type natriuretic peptide (NT-proBNP) is an inactive and stable amino acid fragment associated with LVEF, secreted by cardiac myocytes in response to neurohormonal stimulation, transmural wall stress, inflammation, and ischemia [3, 4]. Elevated levels of NT-proBNP indicate left ventricular remodeling, which is associated with major adverse cardiovascular events (MACE), reduced quality of life, an increased risk of rehospitalization, and higher cardiovascular mortality rates [5].

Modified EAT-Lancet diet

Overweight and obesity, characterized by excessive fat accumulation [6], are common comorbidities that play a critical role in the pathogenesis of HF through hemodynamic and morphological cardiac changes [7, 8]. Lifestyle interventions, particularly dietary modifications and physical activity, are the cornerstone of any obesity management program, especially among HF patients [9, 10]. To date, extensive evidence has highlighted the benefits of “heart-healthy” dietary patterns, such as the Mediterranean diet (Med diet), the Indo-Mediterranean diet, and the Dietary Approaches to Stop Hypertension diet (DASH), all of which have been consistently proven to be cardioprotective [11–19]. Additionally, diet has a significant influence on both human health and the well-being of the planet [20]. The EAT-Lancet Commission proposed the EAT-Lancet universal healthy reference diet in 2019 as a “Sustainable Diet (SD)”, which not only prevents diet-related diseases but also promotes environmental sustainability [20]. The SD emphasizes the consumption

of whole grains, vegetables, fruits, legumes, nuts, and fish, along with moderate amounts of dairy and poultry, while limiting the intake of red meat, tubers and starchy vegetables, processed and prepackaged foods, added fats, and sugars [21]. Higher adherence to SD has been associated with a reduced risk of HF [21].

Time-Restricted Eating (TRE)

In addition to dietary interventions, the timing of food consumption has been found to play a promising role in cardiometabolic health, independent of weight loss [22]. Intermittent fasting (IF) is a well-tolerated eating pattern involving periods of voluntary abstinence from calories, alternated with periods of calorie consumption [22]. Furthermore, IF can promote a metabolic switch by increasing the utilization of ketone bodies (KBs) during fasting in highly metabolically active organs, such as the heart [23]. IF is a general term for various dietary modalities, including periodic prolonged fasting (e.g., fasting for 2–7 days), alternate-day fasting (ADF) (fasting day alternated with ad libitum consumption day), the 5:2 fasting regimen (fasting for 2 days per week), religious fasting (e.g., Ramadan), and TRE (which limits daily food consumption to a window of 4–12 h, followed by a fasting window of 12–20 h per day) [22, 24]. There is still a knowledge gap regarding the optimal dietary intervention for improving HF. Given this, the combination of TRE with distinct dietary interventions (namely, SD) may improve and reverse cardiovascular events among HFrEF patients who are overweight or obese.

Gut hypothesis and Gut-Heart Axis (GHA)

On the other hand, the “gut hypothesis” of HF proposed that the microbial composition of the gut plays a crucial role in the pathogenesis of HF [25]. Recent studies have shown that dysbiosis of the gut microbiota increases the production of trimethylamine-N-oxide (TMAO), a key metabolite generated by the gut microbiota from dietary choline, betaine, and L-carnitine [25, 26]. Elevated TMAO levels are associated with increased NT-proBNP levels, greater severity of left ventricular dysfunction, left ventricular dilation, poor functional status (as defined by the New York Heart Association [NYHA] classification), and adverse clinical outcomes among HF patients

[27]. Modifying gut microbiota with probiotics may offer a novel and promising approach for the prevention and treatment of cardiac remodeling [28].

Probiotics

Long-term treatment with *Lactobacillus helveticus* (*L. helveticus*) has been shown to effectively reduce systolic and diastolic blood pressure in patients with hypertension [29]. Additionally, administration of *L. plantarum* 299v (*Lp299v*) improves endothelium-dependent vasodilation and reduces systemic inflammation in patients with stable coronary artery disease (CAD) [30]. Moreover, probiotic-fermented purple sweet potato yogurt (PSPY) (containing *L. acidophilus*; *L. delbrueckii*; and *L. gasseri*) demonstrated an inhibitory effect on cardiac hypertrophy in 6-week-old mice [31]. The findings of previous studies on the effects of probiotic administration in HF have been contradictory. For instance, Shoei Matin et al. reported a favorable effect of multi-strain synbiotic therapy on NT-proBNP levels in HFrEF patients [27]. Moreover, *L. rhamnosus* GR-1 attenuates HF in rats [32]. However, another study found no significant difference between the consumption of probiotic-enriched yogurt and regular yogurt in HF patients [33].

Rationale and aim of the study

Probiotics have been recognized as an innovative strategy to enhance the effectiveness of dietary interventions for both weight management and clinical improvements [34]. It should also be noted that the selection of *Lp299v* is based on preclinical and clinical evidence demonstrating its gastrointestinal survival, adherence to the intestinal epithelium, contribution to gut health, and beneficial effects on iron absorption, immune function, and cardiovascular outcomes [30, 32, 35, 36]. To the best of our knowledge, no clinical trial has investigated the effects of co-administration of SD and TRE with probiotic supplementation in HF patients with obesity or overweight. Therefore, the present study aims to assess the effects of co-administration of SD and TRE, either alone or in combination with *Lp299v* supplementation, compared to control group, on anthropometric indices, clinical signs, echocardiographic parameters, arterial function, and biochemical markers in overweight or obese HFrEF patients. In fact, in our study, the combination of diet and *Lp299v* may synergistically exert favorable effects on nutritional and inflammatory biomarkers, and clinical outcomes.

Subjects and methods

Study design

This placebo-controlled, three-parallel-arm, randomized, double-blind clinical trial is designed to investigate the effects of co-administration of SD and TRE, either alone or in combination with *Lp299v* supplementation,

compared to control group, on cardiovascular function in patients with HFrEF. As a double-blind study, the experts responsible for examining patients and measuring laboratory data will be blinded to the intervention agents and group assignments. The study was approved by the Medical Ethics Committee of Mashhad University of Medical Sciences (IR.MUMS.REC.1402.259; approved on December 16, 2023). Additionally, the study protocol has been registered with the Iranian Registry of Clinical Trials (IRCT20231225060520N1; registered on January 23, 2024). This trial will be conducted at Imam Reza Hospital, Mashhad, Iran. Written informed consent will be obtained from all participants before their involvement in the study. The study protocol was prepared in accordance with the Standard Protocol Items: Recommendations for Interventional Trials (SPIRIT) 2025 checklist (Additional file 1). The study flow chart for enrolment, allocation, follow up, and analysis is presented in Fig. 1. The SPIRIT diagram for the trial schedule is shown in Fig. 2. Any amendments to the study protocol regarding patient safety, benefits, protocol deviations, or unintentional modifications that do not affect participants' rights, study risk or benefit, data integrity, or overall safety and welfare will be implemented after review and approval by the Department of Clinical Nutrition and the Medical Ethics Committee of Mashhad University of Medical Sciences (MUMS), before the study's initiation. Any changes to the study protocol will also be submitted to the Nutrition Journal (<https://nutritionj.biomedcentral.com/>).

Study duration

The follow-up period will be 60 days. The 60-day duration is set based on evidence from prior nutritional interventions in HF patients [37–39], validation of NT-proBNP as a surrogate endpoint within this timeframe [40, 41], alignment with routine clinical care, and considerations of feasibility and adherence to the intervention.

Study setting

Participants will be recruited from the Cardiac Outpatient Clinic at Imam Reza Hospital, affiliated with Mashhad University of Medical Sciences. Selection will include HFrEF patients who have been previously diagnosed by an expert cardiologist based on both echocardiographic indices (LVEF) and the NYHA classification (II and III). Patients expressing interest will be invited to undergo the screening process.

Study participants and enrollment

Before recruitment, an experienced cardiologist specializing in the diagnosis of HFrEF will identify eligible patients based on LVEF and the NYHA functional class. Following diagnosis, eligible patients with HFrEF will be referred to the researcher responsible for patient

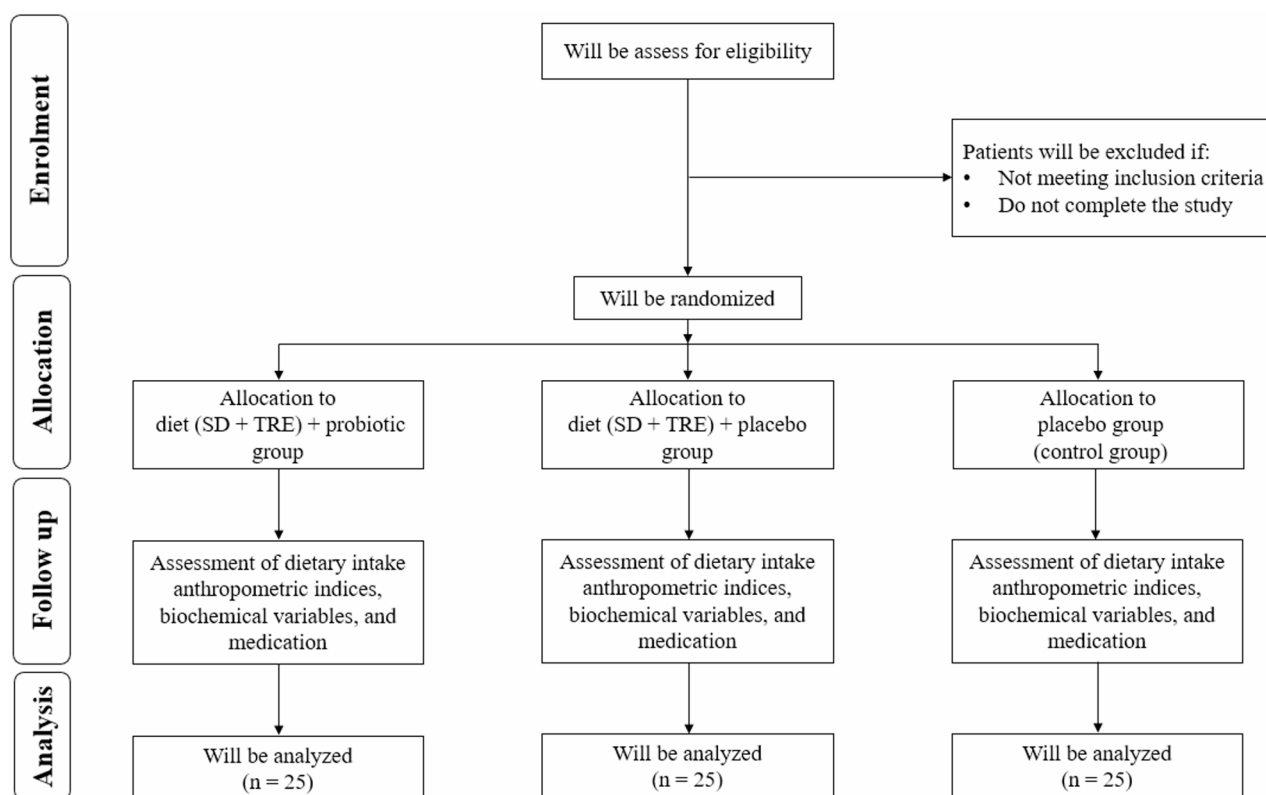


Fig. 1 Study flow chart. SD, Sustainable diet; TRE, Time-restricted eating

selection and enrollment. After receiving a detailed explanation of the study, patients who provide informed consent and meet the inclusion criteria will be enrolled in the study.

Inclusion criteria

The inclusion criteria are as follows: men and women aged 30 to 75 years with an LVEF $\leq 40\%$ (as diagnosed by a cardiologist), who have received standard treatments for HF for at least one month, are classified in NYHA functional class II or III by a cardiologist, and have a BMI between 25 and 35 kg/m².

Non-entry criteria

Patients will not be eligible if they meet any of the following criteria: have experienced an acute cardiac event within the last month; have an intolerance to probiotic supplementation; have an active inflammatory or infectious disease at the time of enrollment; have chronic kidney disease (eGFR ≤ 30 mL/min/1.73 m²) or are receiving renal replacement therapy (dialysis); have liver failure or cirrhosis (Child–Pugh class C); have gastrointestinal diseases (e.g., short bowel syndrome, inflammatory bowel disease) or conditions that affect probiotic absorption (e.g., ileostomy); have cancer; are candidates for cardiac surgery during the study period; are taking antibiotics at the time of enrollment (but may be eligible if treatment is

stopped and the patient recovers after one month); consume fermented products or foods containing probiotics; are unable to fast; are night shift workers; are drug abusers; are on a special or prescribed diet for any reason; are receiving antidiabetic therapy with sulfonylureas or insulin; are participating in other interventional studies; are pregnant or breastfeeding; are reluctant or mentally unprepared to participate in the study.

Baseline eligibility criteria

Participants will be eligible if they do not meet any of the following criteria: have received antibiotics for at least one month before the intervention; known allergy to the probiotic; current use of prohibited supplements/medications (take nutritional supplements or any medication affecting weight, appetite, carbohydrate, or lipid metabolism during the study); and unstable HF therapy.

Post-randomization discontinuation criteria

Post-randomization discontinuation criteria will be: starting antibiotics during the intervention; protocol-defined poor adherence (i.e., consuming less than 80% of the supplements during the study period); adverse reaction to the supplement; systolic blood pressure < 90 mmHg [42]; or disturbances in serum electrolyte levels.

	STUDY PERIOD										
	Enrolment		Post- randomization								Close-out
TIMEPOINT	<i>-t₁ to 0</i>	<i>0</i>	<i>Week 0</i>	<i>Week 1</i>	<i>Week 2</i>	<i>Week 3</i>	<i>Week 4</i>	<i>Week 5</i>	<i>Week 6</i>	<i>Week 7</i>	<i>Week 8</i> ± 5 days
MODE	In-person	In-person	In-person	Phone	Phone	Phone	In-person	Phone	Phone	Phone	In-person
ENROLMENT:											
Eligibility screen	X										
Informed consent	X										
Allocation		X									
INTERVENTIONS:											
<i>Sustainable diet + Time- restricted eating with probiotic supplementation (as first intervention group)</i>											
<i>Sustainable diet + Time- restricted eating with placebo supplementation (as second intervention group)</i>											
<i>Placebo supplementation (as control group)</i>											
ASSESSMENTS:											
<i>Demographic information</i>			X								X
<i>Assessment of quality of life, appetite, sleep quality, depression, anxiety and stress</i>											
<i>Clinical assessment</i>			X				X				X
<i>Concomitant HF medication documentation</i>			X	X	X	X	X	X	X	X	X
<i>Measurement of height (baseline-only)</i>			X								
<i>Measurement of weight, NC, WC, HP, MAC, and body composition</i>			X				X				X
<i>Laboratory tests</i>			X				X				X
<i>Nutritional assessment</i>			X				X				X
<i>Dietary recall</i>			X				X				X
<i>Physical activity assessment</i>			X								X
<i>Pulse wave assessment</i>			X								X
<i>Electrocardiogram and Echocardiography assessment</i>			X								X
<i>Two-minute step test</i>			X								X

Fig. 2 SPIRIT diagram of the recommended content for the schedule of enrolment, intervention, and assessments. Abbreviations: NC, neck circumference; WC, waist circumference; HC, hip circumference; MAC, mid-arm circumference

Study interventions

Participants will be randomly assigned to one of three groups (with an allocation ratio of 1:1:1) for 60 days. The study groups include: (A) co-administration of SD and TRE with probiotic supplementation; (B) co-administration of SD and TRE with placebo supplementation; and (C) a control group with placebo supplementation as the control. In this study, a 14:10 TRE regimen will be implemented, with a 14-hour fasting window followed by a 10-hour eating window. During the eating window, participants will be instructed to follow the SD diet, which will be tailored to each individual by a dietitian. Participants will select one of the following time ranges for their eating period: [1] from 9:00 AM to 7:00 PM daily; [2] from 10:00 AM to 8:00 PM daily; or [3] from 11:00 AM to 9:00 PM daily. The fasting period will be from 7:00 PM to 9:00 AM the following day, from 8:00 PM to 10:00 AM, or from 9:00 PM to 11:00 AM, respectively. During the 14-hour fasting period, participants will only be allowed to drink water, tea, black coffee (without sugar or milk), or any other calorie-free beverage.

SD will be designed based on the EAT-Lancet diet, as follows;

(1) Definition of EAT-Lancet diet

Food systems have the potential to promote human health and support environmental sustainability; however, they are currently threatening both. With the growing global population, providing healthy diets through sustainable food systems is increasingly challenging. Over 820 million people suffer from food insufficiency, and many others adopt unhealthy eating patterns that contribute to diet-related obesity and non-communicable diseases (NCDs), such as diabetes, stroke, HF, and premature death. Given the above, the global burden of NCDs is expected to worsen. At the same time, the environmental impact of food production—on greenhouse gas emissions (GHGE), nitrogen and phosphorus pollution, biodiversity loss, and water and land use—will further destabilize the Earth's system [20].

To ensure alignment with the UN Sustainable Development Goals (SDGs) and the Paris Agreement, the EAT-Lancet Commission convened 19 commissioners and 18 coauthors from 16 countries in 2019, representing diverse fields such as human health, agriculture, political science, and environmental sustainability. The goal was to develop global scientific targets based on the best available evidence for healthy diets and sustainable food production. The proposed reference diet primarily consists of vegetables, fruits, whole grains, legumes, nuts, and unsaturated oils, with a moderate

amount of seafood and poultry, and minimal to no red meat, processed meat, added sugar, refined grains, and starchy vegetables. The global average intake of healthy foods is substantially lower than the reference diet, while overconsumption of unhealthy foods continues to rise. The widespread adoption of the reference dietary pattern would yield significant health benefits, including a substantial reduction in total mortality. Moreover, achieving healthy diets through sustainable food systems for all will require substantial shifts towards healthier dietary patterns, significant reductions in food losses and waste, and major improvements in food production practices [20].

(2) Definition of EAT-Lancet diet index

The food components in the EAT-Lancet diet index [43] are based on the EAT-Lancet diet introduced by Willett et al. in 2019 [20]. The index was previously validated by Stubbendorff et al., based on its association with mortality [43]. The food components of the EAT-Lancet diet are categorized as either “emphasized foods” or “limited foods,” as described in previous accounts of the diet [20, 43]. Emphasized foods include vegetables, fruits, unsaturated oils, legumes, whole grains, nuts, and fish. Limited foods consist of beef and lamb, pork, poultry, eggs, dairy, potatoes, and added sugar [20, 43]. Table 1 provides detailed information about the 14 components of the EAT-Lancet diet.

(3) Calculation of energy and macronutrient distribution

First, the basal metabolic rate (BMR) of each patient will be calculated individually using the Mifflin-St Jeor equation as follows:

$$\begin{aligned} BMR \text{ for men (kcal/day)} \\ &= 10 \times \text{weight (kg)} + 6.25 \times \text{height (cm)} \\ &\quad - 5 \times \text{age (year)} + 5 \end{aligned}$$

$$\begin{aligned} BMR \text{ for women (kcal/day)} \\ &= 10 \times \text{weight (kg)} + 6.25 \times \text{height (cm)} \\ &\quad - 5 \times \text{age (year)} - 161 \end{aligned}$$

Second, the total energy expenditure (TEE) for each individual will be calculated based on their physical activity level (PAL) and the thermic effect of food (TEF). TEF typically accounts for 10% of TEE.

Physical Activity Level (PAL) is categorized into different levels based on daily activity intensity:

- Sedentary (1.2): Little to no physical activity
- Lightly active (1.375): Light exercise or sports 1–3 days per week

Table 1 Criteria for the EAT-Lancet index

Food component in EAT-Lancet diet index ¹		Target intake	Reference interval ²	3 points	2 points	1 point	0 points
Emphasized intake							
	Vegetables	300	200–600	> 300	200–300	100–200	< 100
	Fruits	200	100–300	> 200	100–200	50–100	< 50
	Unsaturated oils	40	20–80	> 40	20–40	10–20	< 10
	Legumes	75	0–150	> 75	37.5–75	18.75–37.5	< 18.75
	Nuts	50	0–100	> 50	25–50	12.5–25	< 12.5
	Whole grains	232		> 232	116–233	58–116	< 58
	Fish	28	0–100	> 28	14–28	–714	< 7
Limited intake							
	Beef and lamb	7	0–14	< 7	7–14	14–28	> 28
	Pork	7	0–14	< 7	7–14	14–28	> 28
	Poultry	29	0–58	< 29	29–58	58–116	> 116
	Eggs	13	0–25	< 13	13–25	25–50	> 50
	Dairy	250	0–500	< 250	250–500	500–1000	> 1000
	Potatoes	50	0–100	< 50	50–100	100–200	> 200
	Added sugar	31	0–31	< 31	31–62	62–124	> 124

This table was adapted from “Development of an EAT-Lancet index and its relation to mortality in a Swedish population”

¹Food components in the index are based on the EAT-Lancet diet as grams per day

²Target and reference values from the EAT-Lancet diet, based on an energy intake of 2500 kcal, expressed in grams

- Moderately active (1.55): Moderate exercise or sports 3–5 days per week
- Very active (1.725): Hard exercise or sports 6–7 days per week
- Super active (1.9): Very intense exercise, physical job, or training twice a day

$$TEE \text{ (kcal/day)} = BMR \times PAL \times TEF$$

Third, this daily caloric requirement will be divided into seven meals, including three main meals and four snacks, with a macronutrient ratio of 50% carbohydrates, 35% fat, and 15% protein. Additionally, these meals will adhere to the EAT-Lancet diet index. A dietitian will carry out all of the above-mentioned steps.

(4) Food intake conversion

In the study by Stubbendorff et al., the 14 food components in the EAT-Lancet index are presented in grams per day [43]. Target and reference values from the EAT-Lancet diet were presented based on an energy intake of 2500 kcal, expressed in grams. For lower calorie levels, we will calculate the ratio. For instance, in the SD with 2000 kcal, we will divide 2000 by 2500 to obtain the ratio of 2000 kcal SD to 2500 kcal SD. This ratio will then be multiplied by the target grams for each food component in the 2500 kcal EAT-Lancet diet. Finally, based on the calculated grams, the 2000 kcal EAT-Lancet diet will be prepared. For other calorie levels, the steps mentioned above will be applied. Since pork is forbidden in Iran based on Islamic religious teachings, it has been replaced with lean red meat

in the design of a SD. Furthermore, in this study, we use the term “SD” as a substitute for the EAT-Lancet diet, with both terms being equivalent. Table 2 presents detailed information about the SD.

- (5) Preparing the individualized food menu and providing detailed explanations to participants Iranian dietary preferences will be taken into account when preparing individualized food menus based on the SD. Additionally, to develop a SD menu, we will incorporate multicultural dietary models, including the Indo-Mediterranean and HEART diets, both of which are rich in antioxidant components [14–17]. Individualized menus will be designed based on these principles, and the intervention will be implemented using a culturally adapted EAT-Lancet framework (modified EAT-Lancet Diet). Additionally, in designing the SD diet, we pay particular attention to biodiversity, namely both inter- and intra-food group diversity, in terms of the variety of plant and animal species. In the SD menu, whole grains will be emphasized (e.g., incorporating 50% wheat porridge into wheat flour for bread preparation), while refined grains will be completely excluded. To reduce the glycemic index, the addition of bran to rice prior to consumption will be considered, along with planning combinations of rice with different types of legumes. These measures are in line with reducing the glycemic index, improving biodiversity, and lowering carbon emissions. Moreover, to ensure palatability, commonly used spices in Iranian cuisine will be recommended for incorporation into main meals.

Table 2 Food component of SD and conversion of 2500Kcal SD to 2000 Kcal SD**Calculation of Energy (based on Mifflin-St Jeor) and scaling factor:***BMR for men (kcal/day)* = $10 \times \text{weight (kg)} + 6.25 \times \text{height (cm)} - 5 \times \text{age (year)} + 5$ *BMR for women (kcal/day)* = $10 \times \text{weight (kg)} + 6.25 \times \text{height (cm)} - 5 \times \text{age (year)} - 161$ *TEE (kcal/day)* = *BMR* × *PAL* × *TEF**Scaling factor* = *TEE* / 2500

	Food component of SD ¹	2500 Kcal		Conversion of 2500 Kcal to 2000 Kcal ³	2000 Kcal	
		Target intake	Reference interval ²		Target intake	Reference interval
Emphasized intake				<i>Scaling factor = 0.8</i>		
	Vegetables (g)	300	200–600	300 (g) × 0.8= 240 (g)	240	160–480
	Fruits (g)	200	100–300	200 (g) × 0.8= 160 (g)	160	80–240
	Unsaturated oils (g)	40	20–80	40 (g) × 0.8= 32 (g)	32	16–64
	Legumes (g)	75	0–150	75 (g) × 0.8= 60 (g)	60	0–120
	Nuts (g)	50	0–100	50 (g) × 0.8= 40 (g)	40	0–80
	Whole grains (g)	232		232 (g) × 0.8= 186 (g)	186	
	Fish (g)	28	0–100	28 (g) × 0.8= 22.4 (g)	22.4	0–80
Limited intake						
	Beef and lamb (g) ²	14	0–28	14 (g) × 0.8= 11.2 (g)	11.2	0–22.4
	Processed/pre- served meat	0	0	0 (g) × 0.8= 0 (g)	0	0
	Poultry (g)	29	0–58	29 (g) × 0.8= 23.3 (g)	23.3	0–46.4
	Eggs (g)	13	0–25	13 (g) × 0.8= 10.4 (g)	10.4	0–20
	Dairy (g)	250	0–500	250 (g) × 0.8= 200 (g)	200	0–400
	Potatoes (g)	50	0–100	50 (g) × 0.8= 40 (g)	40	0–80
	Added sugar (g)	31	0–31	31 (g) × 0.8= 24.8 (g)	24.8	0–24.8

Abbreviations: *BMR* Basal metabolic rate, *TEE* Total energy expenditure, *PAL* Physical activity level, *TEF* Thermic effect of food¹Food component of SD presented as gram per day based on Malmö Diet and Cancer Study²Since Iranians, due to being Muslim, consider pork forbidden based on Islamic religious teachings, in the design of a sustainable diet, pork has been replaced with lean red meat³To convert the 2500 Kcal EAT-Lancet diet to other calorie levels (e.g., 1800 kcal, 2000 kcal, 3000 kcal), we need to scale the recommended food groups proportionally. The EAT-Lancet diet provides target intake levels for different food groups, and adjusting them based on a new calorie intake requires maintaining the same macronutrient and food group distribution [For example, 2000 divided by 2500 equals 0.8 (scaling factor)]

The energy and nutrient content of food items will be analyzed using the Iranian Food Composition Table (FCT). USDA FCT may be used only in case of those foods, for which Iranian data is not available.

After preparing the individualized food menu, the researcher will provide a thorough explanation to the participants to ensure understanding and maximize adherence to the dietary intervention. Additionally, to enhance flexibility, cultural acceptability, and adherence to the dietary plan, twelve characteristics of a healthy diet will be emphasized.

Randomization and blinding

Randomization will be performed before participant allocation. A stratified permuted block randomization (with a fixed block size of $n = 4$) will be conducted using computer-generated random numbers by the study methodologist through the online website (<http://www.sealdenvelope.com>). In this method, eligible participants who meet

the study inclusion criteria will be randomly allocated to one of three groups, with stratification by BMI (25–30 kg/m² and 30–35 kg/m²): SD + TRE with a probiotic supplement, SD + TRE with a placebo, and control group with a placebo. Random allocation will be based on a random number list, with the letters A, B, and C assigned to an equal number of participants in the list (the allocation ratio is 1:1:1 for groups A, B, and C).

In this study, for the first time in Iran, we will innovate by producing a specific probiotic, *Lp299v*, for HF patients with a therapeutic dose of 1.4×10^{10} colony-forming units (CFU) per capsule. Both placebo and probiotic supplements will be produced by Pardis Roshd Company (Shiraz, Iran). The packages of the capsules and the capsules will be identical in terms of color, flavor, and taste to conceal the intervention and placebo groups. The packages will then be placed in pharmaceutical polyethylene bottles and assigned a special code. The codes for both placebo and probiotic supplements will be retained by the company until the end of the study, ensuring blinding for researchers, laboratory staff, outcome assessors, and patients. However, no blinding will be implemented

concerning the dietary intervention (SD + TRE or control group). We do not anticipate the need for unblinding; however, if necessary, the trial manager, data coordinator, and implementation support facilitators will have access to group allocations, and any unblinding will be documented.

Supplement administration

During the study period, participants will take two probiotic or placebo supplement capsules daily—one an hour after breakfast and the other two hours after dinner, accompanied by a glass of water. Each probiotic capsule contains 1.4×10^{10} CFU of *Lactobacillus plantarum* 299v (*Lp299v*), along with Avicel®, magnesium stearate, and colloidal silica. The placebo capsule contains croscarmellose, Avicel®, mannitol, colloidal silica, and flavoring, but no probiotic.

Probiotic selection rationale

Among the diverse probiotic genera commonly utilized in dietary supplements and functional foods, *Lactobacillus* represents one of the most extensively studied and clinically relevant groups. For instance, a preclinical study demonstrated that administration of *L. rhamnosus* GR-1 attenuates myocardial hypertrophy and HF following myocardial infarction, suggesting that probiotics may hold promise as a potential therapeutic strategy for improving HF (probiotic-mediated cardiac improvement) [32]. While several species are adapted to specific ecological niches, *L. plantarum* is widely distributed across dairy products, meat, fermented foods, and vegetables. The strain *Lp299v* is recognized as safe and has been applied in both food and pharmaceutical contexts, with reported health benefits in a range of conditions, including neurodegenerative and metabolic disorders, cancer, urogenital and hepatic diseases, anemia, and hypertension [35, 36].

Given the broad cardiovascular therapeutic potential of probiotics [35], it is important to elucidate the strain-specific benefits and characteristics of *Lp299v* in the context of HFrEF. However, evidence-based guidance regarding the optimal strain and dosage of probiotic supplementation in patients with HFrEF is currently lacking. Therefore, we selected this specific strain of *Lp299v* and its dosing regimen based on previously mentioned studies [32, 35, 36] and also the study that demonstrated the favorable impact of *Lp299v* supplementation on improved vascular endothelial function, reduced circulating levels of interleukins 8 and 12, and increased propionic acid production among men with stable coronary artery disease (CAD) [30].

Adherence to the intervention

All participants will receive weekly telephone calls and be scheduled for monthly meetings with the research team during the intervention. These calls and meetings will ensure comprehension of the recommendations, encourage adherence, and remind participants of the study protocol. Furthermore, all participants (intervention groups) will receive detailed written and oral instructions for the interventions. Participants in the SD + TRE with probiotic supplementation group and the SD + TRE with placebo group will receive a diet plan for each meal and snack tailored to their daily energy requirements. Participants in the control group does not receive, nor have reemphasis on, verbal advice regarding food intake, in order to allow measurable differences in adherence between the intervention and control groups. Monitoring and follow-up procedures will be consistent among all participants, regardless of group allocation.

As mentioned above, there are no differences between patients in each intervention group or the control group in terms of monitoring and follow-up. All participants will be provided with the researcher's contact number and encouraged to reach out with any questions. Similarly, the researcher will contact participants to assess compliance with the personalized diet (intervention groups), track the consumption of intervention capsules (all groups), through phone conversations.

Participants will receive standardized daily reminders through instant messages, clearly indicating the prescribed start and end times of their eating window (SD + TRE with probiotic supplementation group and SD + TRE with placebo group). To ensure adherence, dietary monitoring will be carried out through structured daily interactions between the researcher and each participant, primarily conducted via scheduled telephone calls. These interactions will allow the researcher to provide guidance, address potential challenges, and verify compliance with the dietary protocol. Participants will be considered non-compliant if they miss more than two consecutive phone sessions, consume ≥ 3 inappropriate meals (e.g., consumption of fast foods, simple sugars exceeding the recommended allowance in the prescribed diet, animal fats/saturated fats, organ meats, processed meat products, or bread made from refined white flour), or fail to follow the respective diet instructions ≥ 3 times a week for more than two consecutive weeks. Participants demonstrated a high compliance rate ($> 90\%$), defined as consumption of more than 85% of the prescribed menu [44, 45]. A 3-day, 24-hour dietary recall will be collected from participants on days 0, 30, and 60 to confirm compliance, too.

Furthermore, capsule count will be done in the containers supplied to patients, during each visit to find out the adherence of probiotics.

Outcomes

Primary outcomes

The primary outcomes of this study will be the measurement of NT-proBNP and LVEF. NT-proBNP concentrations will be determined using an electrochemiluminescence immunoassay (Elecsys® proBNP II, Roche Diagnostics). The ECLIA method will be performed on the automated cobas e 411 immunoassay analyzer. A predefined acceptable coefficient of variation (CV) of <5% will be applied to ensure the precision of the assay.

Secondary outcomes

- Cardiometabolic biomarkers will include complete blood count (CBC), fasting blood glucose (FBS), lipid profile (total cholesterol [TC], low-density lipoprotein cholesterol [LDL-c], high-density lipoprotein cholesterol [HDL-c], and triglycerides [TG]), liver function markers (aspartate aminotransferase [AST], alanine aminotransferase [ALT], bilirubin total, bilirubin direct, and alkaline phosphatase [ALP]), kidney function markers (creatinine [Cr], and urea), uric acid, sodium (Na), potassium (K), albumin (Alb), zonulin, TMAO, and cardiac troponin.
- Anthropometric indices will encompass weight (W), height (H), body mass index (BMI), waist circumference (WC), hip circumference (HC), waist-to-hip ratio (WHR), neck circumference (NC), body shape index (ABSI), mid-arm circumference (MAC), fat-free mass (FFM), fat mass (FM), and body water (BW).
- Arterial and additional echocardiographic indices will contain arterial stiffness (pulse wave velocity [PWV], augmentation index [AI], brachial systolic pressure [BSP], brachial diastolic pressure [BDP], brachial pulse pressure [BPP], aortic systolic pressure [ASP], aortic diastolic pressure [ADP], aortic pulse pressure [APP], aortic augmented pressure [AAP], heart rate [HR], systolic blood pressure [SBP], and diastolic blood pressure [DBP]), and echocardiographic indices (left ventricular end-diastolic volume [LVEDV], left ventricular end-systolic volume [LVESV], diastolic dysfunction [DD], and the ratio of early diastolic mitral inflow velocity to early diastolic mitral annulus velocity [E/e' ratio]).
- Functional and patient-reported outcomes will incorporate sleep quality, appetite, energy intake, quality of life, DASS-21, and exercise tolerance (measured via the 2-minute step test).

All of the above outcomes will be evaluated both before and after the intervention. Furthermore, endpoint

collection will be conducted by third-party assessors who are blinded to group allocation.

Cardiac troponin, FBS, lipid profiles, creatinine, echocardiographic indices, and body weight will be prespecified as key secondary outcomes. Additionally, zonulin and TMAO will be assessed as exploratory endpoints.

Measurements and assessments

Demographic characteristics Demographic characteristics of patients, including age, level of education, occupation, marital status, duration of disease diagnosis, sleep time (time to sleep and time to wake up), alcohol consumption, smoking history, use of nutritional supplements, use of herbal medicines, and history of various diseases, will be collected at baseline using a checklist during an interview.

Nutritional assessment To assess food intake, a 3-day, 24-hour dietary recall (including two working days and one weekend day) will be obtained through an interview with a trained nutritionist. Participants will be asked to accurately describe their consumption of food, alcoholic, and non-alcoholic beverages over the previous three days (two working days and one weekend day). The interviewer will inquire about meal times, types, and serving sizes of food, as well as the ingredients of the consumed food items. Additionally, dietitians would use food measures, food portions and food models to assess food intakes. The 24-hour dietary recall for the first day will be completed during the first visit, and the trained nutritionist will conduct the second and third recalls via phone. The 24-hour dietary recalls will be collected on a monthly basis. After each recall, the dietary data will be reviewed by a nutrition expert. Dietary intake will be analyzed using Nutritionist IV software (First Databank, San Bruno, CA, USA), modified for Iranian foods.

Physical activity assessment We will assess physical activity using the long-form International Physical Activity Questionnaire (IPAQ) through face-to-face interviews. The validity and reliability of the IPAQ have been previously established in the Iranian population [46]. The IPAQ evaluates habitual physical activity (PA) over the past seven days and consists of 27 items across four PA domains: occupational (7 items), transportation (6 items), household/gardening (6 items), and leisure-time activities (6 items). Additionally, it includes two questions about the time spent sitting, serving as an indicator of sedentary behavior. The questionnaire records the number of days per week and the duration per day spent walking, as well as engaging in moderate- and vigorous-intensity activities across all four domains. To improve accuracy, culturally relevant examples of moderate- and vigorous-

intensity activities are provided. IPAQ data are converted into metabolic equivalent (MET) scores (MET-min/week) by multiplying the time spent on each activity by its corresponding MET value. One MET represents the energy expenditure at rest, approximately 3.5 ml O₂/kg/min in adults. Based on the IPAQ scoring system, physical activity levels are categorized into three groups: inactive, minimally active, and health-enhancing physically active. Additionally, sufficient vigorous activity is defined as engaging in vigorous-intensity exercise for at least 20 min per day on three or more days per week. Similarly, sufficient moderate and walking activities are defined as performing moderate-intensity exercise or walking for at least 30 min per day on five or more days per week [47].

Quality of life assessment In the present study, we will use the reliable and validated Minnesota Living with Heart Failure Questionnaire (MLHFQ) to assess how HF impacts various aspects of a patient's life [48]. This questionnaire focuses on six domains: signs and symptoms, social well-being, physical activity, sexual health, work, and emotional well-being. The MLHFQ is a self-report instrument that employs a Likert-type response scale, allowing respondents to indicate the level of agreement or impact they experience. The scale ranges from 0 (indicating no effect on quality of life) to 5 (indicating the highest level of influence on quality of life). In this context, higher scores indicate poorer quality of life, suggesting that the individual is experiencing significant challenges related to their health condition.

Appetite assessment Appetite will be assessed using the original version of the Council on Nutrition Appetite Questionnaire (CNAQ), developed by Wilson et al. [49]. This questionnaire demonstrates strong psychometric properties and is suitable for measuring appetite in patients with HF [50]. The CNAQ includes eight items that evaluate different aspects of appetite: general appetite (item 1), satiety (item 2), hunger (item 3), taste perception (items 4 and 5), quantity of food intake (item 6), nausea (item 7), and mood (item 8). Each item is rated on a five-point Likert scale [1–5], with total scores ranging from 8 to 40. Lower scores indicate reduced appetite, and a CNAQ score of ≤ 28 suggests poor appetite and an increased risk of significant weight loss over six months. A shorter version of the questionnaire, the Simplified Nutritional Appetite Questionnaire (SNAQ), includes four items from the CNAQ (items 1, 2, 4, and 6) and yields a total score ranging from 4 to 20. A SNAQ score of ≤ 14 also indicates poor appetite and a risk of notable weight loss within six months [49]. Both the CNAQ and SNAQ will be used in the present study.

Depression, anxiety, and stress assessment The DASS-21 (Depression, Anxiety, and Stress Scale) questionnaire [51] will be used to quickly detect potential risks related to the severity of core symptoms of depression, anxiety, and stress. The questionnaire consists of 21 items and is a shortened version of the DASS-42. Each scale measures specific characteristics, as outlined below:

- The depression scale evaluates emotional and cognitive symptoms such as dysphoria, anhedonia, hopelessness, self-deprecation, a lack of interest or engagement in life, and an overall sense of life devaluation.
- The anxiety scale assesses signs of autonomic arousal, situational anxiety, and subjective experiences of anxiousness.
- The stress scale evaluates symptoms of chronic, non-specific arousal, including difficulty relaxing, heightened nervous tension, and tendencies to become easily upset, irritable, overreactive, or impatient.

Each item is rated on a 4-point Likert scale, reflecting the extent to which the statement applied to the respondent over the past week:

- 0 = Did not apply to me at all (never).
- 1 = Applied to me to some degree or some of the time (sometimes).
- 2 = Applied to me to a considerable degree or a good part of the time (often).
- 3 = Applied to me very much or most of the time (almost always).

Each of the three DASS-21 scales contains seven items, which are divided into subscales with similar content. Scores for depression, anxiety, and stress are calculated by summing the scores of the relevant items:

- D (depression), including: 3, 5, 10, 13, 16, 17, and 21.
- A (anxiety), including: 2, 4, 7, 9, 15, 19, and 20.
- S (stress), including: 1, 6, 8, 11, 12, 14, and 18.

To calculate the final score, each DASS-21 subscale score must be multiplied by 2. The resulting scores for depression, anxiety, and stress are then categorized into five severity levels: normal, mild, moderate, severe, and extremely severe, as follows:

- “Normal” is defined as D: 0–9, A: 0–7, S: 0–14.
- “Mild” severity is defined as D: 10–13, A: 8–9, S: 15–18.
- “Moderate” severity is defined as D: 14–20, A: 10–14, S: 19–25.

- “Severe” is defined as D: 21–27, A: 15–19, S: 26–33.
- “Extremely severe” is defined as D: 28+, A: 20+, S: 34+.

Sleep quality assessment Sleep disturbance will be assessed using the reliable and validated Pittsburgh Sleep Quality Index (PSQI), developed by Buysse et al. in 1989 [52]. The questionnaire comprises 18 items that contribute to the scoring of seven components: subjective sleep quality, sleep latency, sleep duration, habitual sleep efficiency, sleep disturbances, use of sleep medication, and daytime dysfunction. The total PSQI score ranges from 0 to 21, with higher scores indicating poorer sleep quality. A score of ≤ 5 reflects good sleep quality, whereas a score > 5 denotes poor sleep quality.

Anthropometric assessment The body weight of the participants will be assessed using a bioelectrical impedance analyzer (BIA; InBody 770; Smitch Pte Ltd; USA) to the nearest 100 g while wearing light clothing and standing barefoot. Height will be measured with a stadiometer to the nearest 0.5 cm, with the subject standing barefoot and with shoulders in normal alignment. BMI will be calculated as weight (in kilograms) divided by the square of height (in meters). Waist, hip, and neck, as well as mid-arm circumference (MAC), will be measured to the nearest 0.5 cm using a measuring tape, over light clothing, without applying pressure. The body shape index (ABSI) will be calculated as follows: $(WC)/(BMI^{2/3} \times Height^{1/2})$.

Body composition assessment Body fat mass (FM) and fat-free mass (FFM) will be measured using a bioelectrical impedance analyzer (BIA; InBody 770; Smitch Pte Ltd; USA). During measurement, participants will remove their shoes and socks, wear light clothing, and stand with the soles of their feet in contact with the foot electrodes while holding the hand electrodes in their bare hands.

Clinical assessment

NYHA classification In the present study, the NYHA classification will be assessed at baseline and after the 60-day intervention by a cardiologist. The cardiologist assigns participants to one of four functional classes, ranging from no physical limitation during activity (class I) to symptoms present at rest (class IV). Participants with an NYHA classification of class II or III at the study's start will be included.

Edema The cardiologist will assess edema using the pitting test, which involves gently pressing a finger on the medial malleolus, the bony portion of the tibia, and the dorsum of the foot for 5 to 15 s. After releasing the pressure, a dimple (pit) will appear in the skin, indicating the presence and extent of edema. The edema grading scale

measures the rebound after the pitting test and is as follows: Grade 1: immediate rebound with a 2 mm pit (+1); Grade 2: rebound in less than 15 s with a 3 to 4 mm pit (+2); Grade 3: rebound greater than 15 s but less than 60 s with a 5 to 6 mm pit (+3); and Grade 4: rebound between 2 and 3 min with an 8 mm pit (+4).

Concomitant HF medication documentation

HF medications will be recorded in detail at baseline as concomitant medications, and any changes (including dosage reduction or removal of drugs) during the trial will be documented. Reduction or discontinuation of medication due to the intervention will be considered a marker of its efficacy. Additionally, we focus adherence monitoring on the probiotic/placebo and dietary intervention, not on routine medical therapy.

Biochemical assessment

For biochemical assessment, 10 mL of venous blood will be drawn after 11–13 h of overnight fasting and transferred to the research laboratory for analysis. The serum will be separated from the blood samples by centrifugation at 3500 rpm for 10 min at 25 °C, then moved to sterile microtubes and stored at –20 °C until analysis. The following parameters will be measured: CBC, FBS, TC, LDL-c, HDL-c, TG, AST, ALT, UA, Cr, BUN, Na, K, Alb, total and direct bilirubin (Bili D, T), ALP, and NT-proBNP. Any remaining samples will be stored at –80 °C for future use at Mashhad University of Medical Sciences.

Arterial stiffness assessment

An ultrasound evaluation of the carotid artery will be performed in the supine position, with the head turned 45° away from the side being scanned. Pulse wave velocity (PWV) and blood pressure measurements will be conducted using the Sphygmocor XCEL device on the right side of each participant. Blood pressure will be measured after at least 15 min of rest on the right upper arm. Additionally, the augmentation index (AIx), central blood pressure (cBP), and heart rate (HR) will be measured according to the manufacturer's guidelines for pulse wave analysis. Since AIx is inherently dependent on heart rate, the AIx75, which standardizes AIx to a heart rate of 75 beats per minute, will also be used. Furthermore, arterial stiffness will be assessed using a tonometer at the femoral and carotid arteries to calculate carotid-femoral PWV. PWV will be examined using electrocardiogram-gated left carotid and radial waveforms (applanation tonometry) measured sequentially with the stepwise method. The travel distance of the pulse wave will be calculated as the distance from the suprasternal notch to the femoral artery and from the suprasternal notch to the carotid artery, using a flexible tape measure. Measurements will be performed at least twice, and the mean PWV will be

used in the analysis. PWV will be measured at baseline and the end of the intervention.

Electrocardiogram (ECG)

All ECGs will be performed in accordance with the standard operating procedure. In brief, patients will be in a resting, supine position for at least 5 min before the ECG is recorded, with the electrodes placed in the standard anatomical positions. During the measurements, patients will be instructed to remain still and relaxed. The recorded 12-lead ECGs will be analyzed using a medical ECONET device. The PR interval will be measured as the time from the beginning of the P wave to the beginning of the QRS complex; the QT interval will be calculated from the beginning of the QRS complex to the end of the T wave; and the Tp-e interval will be measured from the peak to the end of the T wave. The QTc interval will be calculated using Bazett's formula [$QTc = QT / \sqrt{(RR \text{ interval})}$]. P wave dispersion will be calculated as the absolute difference between the maximum and minimum P wave durations, QT dispersion as the absolute difference between the maximum and minimum QT intervals, and T wave dispersion as the absolute difference between the maximum and minimum T wave durations. The frontal QRS-T angle will be calculated as the absolute difference between the QRS axis (ventricular depolarization) and T axis (ventricular repolarization). If the result is greater than 180° , it will be corrected by subtracting it from 360° . Additionally, the QRS duration will be measured.

Echocardiography assessment

Echocardiography will be performed to evaluate the mechanisms related to standard cardiac structure and function assessment. Transthoracic cardiac ultrasound will be conducted using a Samsung Medison SonoAce R7 Ultrasound Machine. A single expert cardiologist will be blinded to intervention/control groups and perform all measurements, and all images will be analyzed by one operator who is blinded to the groups, too. The parameters to be measured include: LVEF, left ventricular end-diastolic volume (LVEDV), left ventricular end-systolic volume (LVESV), diastolic dysfunction (DD), and the ratio of early diastolic mitral inflow velocity to early diastolic mitral annulus velocity (E/e' ratio), as a marker of left ventricular diastolic pressure.

Exercise tolerance assessment

In the present study, exercise tolerance of the participants will be assessed using the 2-Minute Step Test (2MST). To perform the 2MST, the number of steps completed in place within two minutes will be recorded. Upon hearing the "start" command, participants will begin marching in place, raising their knees to the appropriate height (midway between the patella and iliac crest). Participants will

be instructed to complete as many steps as possible while ensuring their knees reach the predefined height within a two-minute time frame. Before the actual test, participants will be allowed to practice a few steps to refine their technique and confirm their ability to complete the task. For safety, participants will be allowed to hold onto a stable table to maintain balance during the test. Participants are free to march at their own pace, with the option to slow down or pause if necessary, before resuming until the test is completed. The researcher will record the number of completed steps, inform participants of the remaining time, and encourage them to achieve their best possible performance. The final test result will be the number of steps in which the right knee reaches or exceeds the predefined height [53].

To facilitate a large number of assessments—including anthropometric measurements, biochemical tests, echocardiography, arterial stiffness, ECG, physical function evaluations, and multiple questionnaires—these procedures will be coordinated with participants' routine clinical care visits, such as cardiology follow-ups, echocardiography appointments, and scheduled blood tests. This strategy minimizes additional visits, travel, and time burden, thereby enhancing recruitment feasibility, adherence to the intervention, and would also strengthen the trial's pragmatic relevance.

Sample size

To the best of our knowledge, no similar study has examined the effect of a diet supplemented with probiotics on NT-proBNP levels and LVEF in HF patients. We calculated the sample size based on both primary outcomes, and the larger estimate was determined as the final sample size. Therefore, the final sample size was calculated based on the study by Giuseppe Cocco and David Chu, which reported mean $\pm$ SD values for NT-proBNP of 1537 ± 377 (pg/ml) and 1853 ± 354 (pg/ml) in the intervention (Calorie-restricted diet) and control groups, respectively [54]. The Sample size was calculated using G*Power (v3.1), assuming a mean difference of -316 pg/ml in NT-proBNP, $\alpha = 0.05$, power = 80%, and expected 20% dropout rate, yielding 25 participants per group, based on the following equation:

$$n = \frac{\left(z_{1-\frac{\alpha}{2}} + z_{1-\beta} \right)^2 (S_1^2 + S_2^2)}{\Delta^2}$$

The calculated sample size was 25 participants in each group. This sample size is designed to detect a significant reduction in NT-proBNP in the leading intervention group (co-administration of SD + TRE with probiotic supplementation) compared to the placebo group at the end

of the 60-day intervention period. Additionally, we define a clinically relevant >20% reduction in NT-proBNP levels and/or a >6.7% increase in LVEF as the basis for powering our study.

Data management

Data will be collected at baseline, mid-study, and post-intervention phases. Participant information will be recorded using case report forms (CRFs). Once collected, the data will be entered and managed in an electronic database. Three data collectors will be responsible for gathering the clinical raw data, which will then be submitted to the project supervisor for approval. All documentation will be securely stored in a locked drawer, accessible only to the principal investigator. Each form will contain a unique identification number for each participant, without any personally identifiable information. A clinical trial monitor will periodically oversee the study to ensure patient rights and well-being are protected, protocol adherence is maintained, ethical standards and regulations are followed, essential documentation is available, and collected data are accurately recorded. One of the investigators will be responsible for verifying data coding, security, and storage. Additionally, she will conduct a double-check of data entry and values to ensure accuracy. If a participant reports an adverse event, further information will be required to assess the need for exclusion from the trial. In such cases, unblinding may be permitted based on the criteria set by the Medical Ethics Committee. Mashhad University of Medical Sciences will oversee the ethical review of the data. Participants who discontinue the study or deviate from the protocol will receive a summary of all outcome data. Missing data will be addressed using modern imputation methods for handling dropouts.

Data monitoring and oversight

Given the minimal risk associated with the intervention and measurement procedures, this trial is classified as low-risk. To ensure participant safety, an independent Safety Officer—who is independent from the study team and the clinical care of participants—will oversee safety monitoring of the study. Safety assessments will be conducted at baseline, 30 days, and 60 days by the independent Safety Officer. In accordance with institutional ethics regulations, the researcher will document all adverse effects, as well as any participant-reported complaints associated with the use of the intervention. The adverse effects will then be shared with a team of researchers and physicians. The Ethics Committee will be informed about the adverse effects and will decide whether to discontinue the intervention for the participant or stop the study in terms of serious adverse effects or repeated adverse effects in participants.

Statistical analysis

All analyses will be conducted using the Intention-to-Treat (ITT) method. Thus, all participants who entered the study (regardless of whether they completed the 60-day study period) will be included in the analysis. Missing data will be replaced by multiple imputations. Statistical analyses will be performed using the Statistical Package for the Social Sciences (SPSS) software, Version 24 (SPSS Inc., Chicago, IL, USA). A Q-Q plot and the Shapiro-Wilk test will be applied to assess the normality of the continuous variables. Also, continuous variables will be presented as mean $\pm$ SD or median (25–75 interquartile range) for normally distributed or skewed data, respectively, and categorical variables will be presented as percentages. Additionally, the chi-square test will be used to compare categorical variables. The One-way ANOVA or Kruskal-Wallis test will be applied to compare continuous variables between groups. In contrast, the paired sample t-test or Wilcoxon rank-sum test will be used for within-group comparisons. To compare differences in primary and secondary outcomes between the three study groups at the end of the trial and adjust for potential confounders, the mixed linear effects model will be applied. Furthermore, between-group differences will be assessed by ANCOVA adjusting for baseline values. Additionally, we plan to use regression models to adjust for potential confounding variables that may differ across the three groups. In addition, subgroup analyses will be conducted based on diabetic status and BMI (if heterogeneity exists). To correct for multiple comparisons (among the predefined secondary outcomes) and reduce the risk of Type I error, p-values will be calculated using the Bonferroni method (Bonferroni correction). Moreover, A Per-Protocol (PP) analysis will also be undertaken to complement and contrast the Intent-to-Treat (ITT) results, and will serve as a sensitivity analysis. Statistical significance will be defined as p -values <0.05.

Consent

This study explicitly state that the informed consent process will be conducted in compliance with Good Clinical Practice (GCP) guidelines. This will include confirmation that participants will receive adequate verbal and written information about the study, have the opportunity to ask questions, and provide voluntary written consent before any study-specific procedures are undertaken.

Biological specimens will also be collected for storage during this trial.

Discussion

HF represents a significant public health challenge, with high morbidity and mortality rates [7]. Overweight and obesity can be seen as a “perfect storm” for cardiovascular disease (CVD), serving as a leading cause of HF [55].

Furthermore, due to its metabolic and inflammatory abnormalities associated with obesity, obesity further complicates HF treatment, particularly in patients with HFrEF [56, 57]. Additionally, disruptions in circadian rhythm, poor diet, intestinal microbial dysbiosis, and fluctuations in NT-proBNP levels can play pivotal roles in the pathogenesis of HF. Moreover, the function of the gut is fundamentally altered in patients with HF. Diminished intestinal blood flow, resulting from decreased cardiac output, correlates with both the aggravation of gut dysbiosis and the severity of HF [58]. These findings have emerged GHA, defined as a bidirectional connection between gut microbiota and cardiovascular health [59], as a key area of research for diagnosing and treating the HF population [58]. While the management of both HF and obesity is interconnected on multiple levels, most recent HF clinical trials have focused on medications and devices and neglected lifestyle modifications, dietary interventions, or the modification of intestinal microbiota [57, 60]. To the best of our knowledge, no clinical trial has examined the effect of combining an IF diet with probiotic supplementation in patients with HFrEF. We hypothesized that the co-administration of SD + 10-hour TRE, alone or in combination with probiotic supplementation, in overweight and obese patients with HFrEF would lead to weight loss and improvements in metabolic, hormonal, and echocardiographic markers compared to control individuals.

A growing body of research has demonstrated that diet has a significant influence on the GHA, a key factor in determining cardiovascular health [61, 62]. For example, previous studies have demonstrated the favorable effects of IF as an intervention strategy, including the restoration of gene expression involved in the body's circadian clock [63], improvement in intestinal microbial composition, regulation of metabolism [63], as well as reductions in blood pressure [64], body weight [64], body fat index [64], fasting blood glucose [65], and lipid oxidation [65], as well as a reduction in insulin resistance [66]. In addition, IF improves walking speed [67], mental and physical function [67], and sleep quality, while reducing inflammation markers such as TNF- α and IL-1 β [65]. The INTERFAST-MI Trial, conducted among patients with ST-segment elevation myocardial infarction (STEMI), revealed that TRE (8:16) was safe and improved left ventricular (LV) function over 4 weeks, with further improvements observed up to 6 months post-myocardial infarction (MI) [42]. Compared with a diet without a fixed fasting regimen, TRE also reduced diastolic blood pressure and BMI [42]. In the mentioned study, TRE reduced serum NT-proBNP concentrations compared to the control group, but this difference was not statistically significant [42].

In addition to the timing of eating, the types of food consumed have received considerable attention in recent studies. Frequent consumption of ultra-processed foods is associated with excessive calorie intake, weight gain, metabolic syndrome, cancer, and cardiovascular diseases [68]. High-fat diets, on the other hand, are linked to changes in the composition of gut microbiota towards the predominance of pro-inflammatory bacteria to beneficial bacteria [61]. In contrast, diets rich in fiber, plant-based foods, and prebiotics promote the growth of beneficial gut bacteria, such as *Lactobacilli* spp., which have been associated with positive cardiovascular outcomes [69]. Furthermore, gut bacteria function as a metabolic organ, influencing the host's metabolism and energy balance [70]. The health outcomes of a diet, along with its low environmental impact, affordability, accessibility, and alignment with socio-cultural preferences, should all be considered [71, 72]. Due to these factors, a SD—similar to the Mediterranean diet but with a stronger focus on environmental impact—would be the best choice for intervention programs aimed at promoting sustainable behaviors [73]. SD is linked to the proliferation of butyrate-producing bacteria, such as *Akkermansia muciniphila*, in the gut microbiota that are associated with anti-inflammatory effects, improved regulation of glucose and lipid profiles, increased glucose sensitivity, reduced insulin resistance, and regulation of blood pressure [71, 74].

In a previous study, supplementation with *Saccharomyces boulardii* improved LVEF among HF patients [75]. To the best of our knowledge, only one study examined the effect of synbiotic therapy on NT-proBNP levels in HF patients. This study found that 10 weeks of daily synbiotic administration significantly reduced NT-proBNP levels; however, no significant differences were observed in high-sensitivity C-reactive protein (hsCRP) levels or blood pressure values [27]. Other benefits of probiotics in terms of improving cardiometabolic health can be exerted through supporting weight management, particularly when combined with IF or other calorie-restricted/healthy diets. If successful, this intervention may provide nutritionists in clinical settings with a new eating pattern (SD + TRE), either alone or in combination with probiotic supplements, as a safe and effective treatment for patients with HF.

Limitations

The limitations of this study are addressed as follows: First, while double-blinding is feasible and implemented for the probiotic versus control groups, blinding participants to the dietary intervention (SD + TRE) is inherently unfeasible. Second, this is a single-center study with an estimated sample size, which may limit the generalizability of the findings. Nevertheless, generating

proof-of-concept data is clinically and ethically essential before planning larger-scale interventions. Should the results from this study prove promising, a multicenter trial will be planned to validate the findings. Third, due to financial and practical limitations, 16 S rRNA analysis is not possible to perform, limiting our ability to fully characterize gut microbiota composition. Fourth, due to the lack of evidence from studies with similar interventions and patient populations, the study duration is set at 60 days. This decision was based on findings from previous studies—which, although not directly related to the intervention or patient population—along with consideration of the common duration of lifestyle modification studies to enhance patient adherence. Consequently, it is possible that changes in some outcomes, may not reach clinical significance within this timeframe. Fifth, adherence to the TRE schedule may pose a challenge; to mitigate this, participants will receive regular reminders via phone calls and instant messages indicating the start and end times of their eating window. Finally, loss to follow-up is anticipated across all three study groups; to address this, a 20% dropout rate has been incorporated into the sample size calculation.

Trial status

Iranian Registry of Clinical Trials
IRCT20231225060520N1 (URL: <https://irct.behdasht.gov.ir/trial/74743>). Registered 23 January 2024. The recruitment started on April 4, 2025, and is expected to be completed by February 4, 2026.

Abbreviations

HF	Heart failure
HFrfEF	Heart Failure with reduced ejection fraction
EF	Ejection fraction
LVEF	Left ventricular ejection fraction
NYHA	New York Heart Association
SD	Sustainable diet
Lp299v	Lactobacillus plantarum 299v
IF	Intermittent fasting
TRE	Time-restricted eating
SCFA	Short-chain fatty acids
TMAO	Trimethylamine N-oxide
CFU	Colony-forming unit
BMI	Body mass index
hsCRP	High-sensitivity C-reactive protein
NT-proBNP	N-terminal pro-B-type natriuretic peptide
GHA	Gut-Heart Axis

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s12937-025-01270-x>.

Supplementary Material 1.

Acknowledgements

This study is a protocol for a Ph.D. thesis, which was approved by the School of Medicine, Department of Nutrition, Mashhad University of Medical Sciences (MUMS), Mashhad, Iran (approval number: IR.MUMS.REC.1402.259;

approved on December 16, 2023). Mashhad University of Medical Sciences is acknowledged for its support in designing the study. We would like to thank Pardis Roshd Company (Shiraz, Iran) for providing us with probiotic supplementation and placebo capsules. The sponsor played no part in the study design, collection, management, analysis, and interpretation of data, writing of the report, or the decision to submit the report for publication.

Dissemination policy

The trial results will be communicated to participants, healthcare professionals, and the general public through trial registry entry, a plain language summary, and publication in scientific journals.

Ancillary and post-trial care

Since participants are at minimal risk during the intervention and measurement protocols, this trial is classified as low-risk. Nevertheless, in case of any harm resulting from participation in the trial, participants will receive appropriate medical care, and compensation will be provided in accordance with ethical and legal regulations. Post-trial care will also be offered if needed.

Authors' contributions

P.R., A.A., S.R.S., K.S., M.K.R., M.H.E., and R.R. conceived and designed the study. P.R., S.A., and F.V. contributed to data collection and assembly. P.R. and G.R. wrote the manuscript. All authors read and approved the final manuscript.

Funding

This research will be funded by the Vice Chancellor's Office for Research and Technology at Mashhad University of Medical Sciences (MUMS), and all study stages, including the design of the study, collection, analysis, and interpretation of data, and writing of the manuscript, will be under its supervision.

The Vice Chancellor of Research and Technology is the sponsor of this trial (The Vice Chancellor of Research and Technology, Ghareshi Building, Daneshgah Ave., Mashhad, Iran).

Data availability

The datasets collected and analyzed during the present study are not publicly accessible due to ethical concerns; however, the corresponding author may provide the datasets upon reasonable request.

Declarations

Ethics approval and consent to participate

The current trial received approval from the Medical Ethics Committee of Mashhad University of Medical Sciences, Mashhad, Iran (IR.MUMS.REC.1402.259; approved on December 16, 2023). Prior to participation in the research project, participants will be asked to provide written informed consent. Consent for Publication: It is optional for patients to participate in the current trial after reading the terms and conditions in the Supplementary Information. A new consent form will be collected from the participants if we need to use additional information from the collected data.

Competing interests

The authors declare no competing interests.

Pardis Roshd Company (Shiraz, Iran) had no role. The funder of the study had no role in study design, data collection, data analysis, data interpretation, management, the writing of the report, or the decision to submit the report for publication.

Author details

¹Department of Nutrition, Faculty of Medicine, Mashhad University of Medical Sciences, Mashhad, Iran

²Student Research Committee, Mashhad University of Medical Sciences, Mashhad, Iran

³Imam Reza Hospital, Mashhad University of Medical Sciences, Mashhad, Iran

⁴Department of Community Nutrition, School of Nutritional Sciences and Dietetics, Tehran University of Medical Sciences, Tehran, Iran

⁵Nuclear Medicine Research Center, Mashhad University of Medical Sciences, Mashhad, Iran

⁶Department of Community Medicine, Faculty of Medicine, Mashhad University of Medical Sciences, Mashhad, Iran

⁷Department of Food Science and Technology, Shiraz University, Shiraz, Iran

⁸Deputy of Education of the Nutrition Department, Faculty of Medicine, Azadi Square, Mashhad University of Medical Sciences, Campus of the University, Mashhad 9177948564, Iran

Received: 15 July 2025 / Accepted: 3 December 2025

Published online: 22 December 2025

References

1. Savarese G, Becher PM, Lund LH, Seferovic P, Rosano GM, Coats AJ. Global burden of heart failure: a comprehensive and updated review of epidemiology. *Cardiovasc Res*. 2022;118(17):3272–87.
2. Bozkurt B, Ahmad T, Alexander K, Baker WL, Bosak K, Breathett K, et al. HF STATS 2024: heart failure epidemiology and outcomes statistics an updated 2024 report from the heart failure society of America. *J Card Fail*. 2025;31(1):66–116.
3. Bozkurt B, Coats AJ, Tsutsui H, Abdelhamid M, Adamopoulos S, Albert N, et al. Universal definition and classification of heart failure: a report of the heart failure society of America, heart failure association of the European society of cardiology, Japanese heart failure society and writing committee of the universal definition of heart failure. *J Card Fail*. 2021;27(4):387–413.
4. Lund LH, Pitt B, Metra M. Left ventricular ejection fraction as the primary heart failure phenotyping parameter. *Eur J Heart Fail*. 2022;24(7):1158–61.
5. York MK, Gupta DK, Reynolds CF, Farber-Eger E, Wells QS, Bachmann KN, et al. B-type natriuretic peptide levels and mortality in patients with and without heart failure. *J Am Coll Cardiol*. 2018;71(19):2079–88.
6. World Health Organization. Obesity: preventing and managing the global epidemic. Report of a WHO consultation. *World Health Organ Tech Rep Ser*. 2000;894:i–xii, 1–253.
7. Lembo M, Strisciuglio T, Fonderico C, Mancusi C, Izzo R, Trimarco V, et al. Obesity: the perfect storm for heart failure. *ESC Heart Fail*. 2024;11(4):1841–60.
8. Lee KS, Park D-I, Lee J, Oh O, Kim N, Nam G. Relationship between comorbidity and health outcomes in patients with heart failure: a systematic review and meta-analysis. *BMC Cardiovasc Disord*. 2023;23(1):498.
9. Visco V, Izzo C, Bonadies D, Di Feo F, Caliendo G, Loria F, et al. Interventions to address cardiovascular risk in obese patients: many hands make light work. *J Cardiovasc Dev Dis*. 2023;10(8):327.
10. Lee VY, Houston L, Perkovic A, Barraclough JY, Sweeting A, Yu J, et al. The effect of weight loss through lifestyle interventions in patients with heart failure with preserved ejection fraction—a systematic review and meta-analysis of randomised controlled trials. *Heart Lung Circ*. 2024;33(2):197–208.
11. Diab A, Dastmalchi LN, Gulati M, Michos ED. A heart-healthy diet for cardiovascular disease prevention: where are we now? *Vasc Health Risk Manag*. 2023;237–53.
12. Wickman BE, Enkhmaa B, Ridberg R, Romero E, Cadeiras M, Meyers F, et al. Dietary management of heart failure: DASH diet and precision nutrition perspectives. *Nutrients*. 2021;13(12):4424.
13. Camafort M, Park S-M, Kang S-M. Lifestyle modification in heart failure management: are we using evidence-based recommendations in real world practice? *Int J Heart Fail*. 2023;5(1):21.
14. Singh RB, Dubnov G, Niaz MA, Ghosh S, Singh R, Rastogi SS, et al. Effect of an Indo-Mediterranean diet on progression of coronary artery disease in high risk patients (Indo-Mediterranean Diet Heart Study): a randomised single-blind trial. *Lancet*. 2002;360(9344):1455–61.
15. Singh RB, Fedacko J, Fatima G, Magomedova A, Watanabe S, Elkilany G. Why and how the Indo-Mediterranean diet May be superior to other diets: the role of antioxidants in the diet. *Nutrients*. 2022;14(4):898.
16. Singh RB, Nabavizadeh F, Fedacko J, Pella D, Vanova N, Jakabcin P, et al. Dietary approaches to stop hypertension via Indo-Mediterranean foods, May be superior to DASH diet intervention. *Nutrients*. 2022;15(1):46.
17. Singh RB, Fedacko J, Pella D, Fatima G, Elkilany G, Moshiri M, et al. High exogenous antioxidant, restorative treatment (heart) for prevention of the six stages of heart failure: the heart diet. *Antioxidants*. 2022;11(8):1464.
18. Mohammadifard N, Rahimlou M, Amani-Beni R, Darouei B, Zarepur E, Haghighatdoost F, et al. Mediterranean diet and prime diet quality score are associated with reduced risk of premature coronary artery disease in Iran: a multi-centric case-control study. *Sci Rep*. 2025;15(1):11385.
19. Bagheri M, Nouri M, Homayounfar R, Akhlaghi M. Association between adherence to the mediterranean diet with cardiometabolic risk factors: a cross-sectional study on PERSIAN cohort study in Fasa. *Sci Rep*. 2023;13(1):14870.
20. Willett W, Rockström J, Loken B, Springmann M, Lang T, Vermeulen S, et al. Food in the anthropocene: the EAT–Lancet commission on healthy diets from sustainable food systems. *Lancet*. 2019;393(10170):447–92.
21. Zhang S, Marken I, Stubbendorff A, Ericson U, Qi L, Sonestedt E, et al. The EAT–Lancet diet index, plasma proteins, and risk of heart failure in a population-based cohort. *JACC Heart Fail*. 2024;12(7):1197–208.
22. Adafer R, Messaadi W, Meddahi M, Patey A, Haderbache A, Bayen S, et al. Food timing, circadian rhythm and chrononutrition: a systematic review of time-restricted eating's effects on human health. *Nutrients*. 2020;12(12):3770.
23. Garza-González S, Nieblas B, Solbes-Gochicoa MM, Altamirano J, García N. Intermittent fasting as possible treatment for heart failure. *Curr Vasc Pharmacol*. 2022;20(3):260–71.
24. Small S, Iglesias-Grau J, Gariepy C, Wilkinson M, Taub P, Kirkham A. Time-restricted eating: a novel dietary strategy for cardiac rehabilitation. *Can J Cardiol*. 2023;39(11):S384–94.
25. Matacchione G, Piacenza F, Pimpini L, Rosati Y, Marcozzi S. The role of the gut microbiota in the onset and progression of heart failure: insights into epigenetic mechanisms and aging. *Clin Epigenetics*. 2024;16(1):175.
26. Lupu VV, Adam Raileanu A, Mihai CM, Morariu ID, Lupu A, Starcea IM, et al. The implication of the gut microbiome in heart failure. *Cells*. 2023;12(8):1158.
27. Shoaeei Matin S, Shidfar F, Naderi N, Amin A, Hosseini-Baharanchi FS, Dehnad A. The effect of synbiotic consumption on serum NTproBNP, hsCRP and blood pressure in patients with chronic heart failure: a randomized, triple-blind, controlled trial. *Front Nutr*. 2022;8:822498.
28. Moludi J, Alizadeh M, Davari M, Golmohammadi A, Maleki V. The efficacy and safety of probiotics intervention in attenuating cardiac remodeling following myocardial infarction: literature review and study protocol for a randomized, double-blinded, placebo controlled trial. *Contemp Clin Trials Commun*. 2019;15:100364.
29. Jauhainen T, Rönneback M, Vapaatalo H, Wuolle K, Kautiainen H, Groop P, et al. Long-term intervention with *Lactobacillus helveticus* fermented milk reduces augmentation index in hypertensive subjects. *Eur J Clin Nutr*. 2010;64(4):424–31.
30. Hofeld BC, Puppala VK, Tyagi S, Ahn KW, Anger A, Jia S, et al. *Lactobacillus plantarum* 299v probiotic supplementation in men with stable coronary artery disease suppresses systemic inflammation. *Sci Rep*. 2021;11(1):3972.
31. Lin P-P, Hsieh Y-M, Kuo W-W, Lin Y-M, Yeh Y-L, Lin C-C, et al. Probiotic-fermented purple sweet potato yogurt activates compensatory IGF-1R/PI3K/Akt survival pathways and attenuates cardiac apoptosis in the hearts of spontaneously hypertensive rats. *Int J Mol Med*. 2013;32(6):1319–28.
32. Gan XT, Ettinger G, Huang CX, Burton JP, Haist JV, Rajapurhitam V, et al. Probiotic administration attenuates myocardial hypertrophy and heart failure after myocardial infarction in the rat. *Circ Heart Fail*. 2014;7(3):491–9.
33. Pourrajab B, Naderi N, Janani L, Mofid V, Hajahmadi M, Dehnad A, et al. Comparison of probiotic yogurt and ordinary yogurt consumption on serum Pentraxin3, NT-proBNP, oxLDL, and ApoB100 in patients with chronic heart failure: a randomized, triple-blind, controlled trial. *Food Funct*. 2020;11(11):10000–10.
34. Álvarez-Arriño V, Martín-Peláez S. Effects of probiotics and synbiotics on weight loss in subjects with overweight or obesity: a systematic review. *Nutrients*. 2021;13(10):3627.
35. Aljohani A, Rashwan N, Vasani S, Alkhawashki A, Wu TT, Lu X et al. The health benefits of probiotic *Lactiplantibacillus plantarum*: a systematic review and meta-analysis. *Probiotics Antimicrob Proteins*. 2024:1–20.
36. Neverovskiy A, Chernyavskiy V, Shypulin V, Hvozdetzka L, Tishchenko V, Nechypurenko T, et al. Probiotic *Lactobacillus plantarum* May reduce cardiovascular risk: an experimental study. *ARYA Atheroscler*. 2021;17(4):1.
37. Fabricio CG, Tanaka DM, de Souza Gentil JR, Amato CAF, Marques F, Schwartzmann PV, et al. A normal sodium diet preserves serum sodium levels during treatment of acute decompensated heart failure: a prospective, blind and randomized trial. *Clin Nutr ESPEN*. 2019;32:145–52.
38. Hassanzadeh-Makoui R, Jamei M, Hassanzadeh-Makoui M, Khederlou H. Effects of vitamin D on left ventricular ejection fraction in patients with systolic heart failure: a double-blind randomized clinical trial. *Int J Endocrinol Metab*. 2020;18(3):e103528.
39. Salmani M, Alipoor E, Navid H, Farahbakhsh P, Yaseri M, Imani H. Effect of L-arginine on cardiac reverse remodeling and quality of life in patients with heart failure. *Clin Nutr*. 2021;40(5):3037–44.

40. Schmitt W, Rühs H, Burghaus R, Diedrich C, Duwal S, Eissing T, et al. NT-proBNP qualifies as a surrogate for clinical end points in heart failure. *Clin Pharmacol Ther.* 2021;110(2):498–507.
41. Pascual-Figal DA, Domingo M, Casas T, Gich I, Ordóñez-Llanos J, Martínez P, et al. Usefulness of clinical and NT-proBNP monitoring for prognostic guidance in destabilized heart failure outpatients. *Eur Heart J.* 2008;29(8):1011–8.
42. Dutzmann J, Kefalianakis Z, Kahles F, Daniel J-M, Gufler H, Wohlgemuth WA, et al. Intermittent fasting after ST-segment-elevation myocardial infarction improves left ventricular function: the randomized controlled INTERFAST-MI trial. *Circ Heart Fail.* 2024;17(5):e010936.
43. Stubbendorff A, Sonestedt E, Ramne S, Drake I, Hallström E, Ericson U. Development of an EAT-Lancet index and its relation to mortality in a Swedish population. *Am J Clin Nutr.* 2022;115(3):705–16.
44. Arciero PJ, Edmonds R, He F, Ward E, Gumprecht E, Mohr A, et al. Protein-pacing caloric-restriction enhances body composition similarly in obese men and women during weight loss and sustains efficacy during long-term weight maintenance. *Nutrients.* 2016;8(8):476.
45. Zuo L, He F, Tinsley GM, Pannell BK, Ward E, Arciero PJ. Comparison of high-protein, intermittent fasting low-calorie diet and heart healthy diet for vascular health of the obese. *Front Physiol.* 2016;7:209013.
46. Moghaddam MB, Aghdam FB, Jafarabadi MA, Allahverdipour H, Nikookheslat SD, Safarpour S. The Iranian version of international physical activity questionnaire (IPAQ) in Iran: content and construct validity, factor structure, internal consistency and stability. *World Appl Sci J.* 2012;18(8):1073–80.
47. Vashghani-Farahani A, Tahmasbi M, Asheri H, Ashraf H, Nedjat S, Kordi R. The Persian, last 7-day, long form of the international physical activity questionnaire: translation and validation study. *Asian J Sports Med.* 2011;2(2):106.
48. Heravi-Karimooi M, Bandari R, Eskandari S, Semnani S, Rejeh N, Montazeri A. Validation study of the Iranian version of Minnesota Living With Heart Failure Questionnaire (MLHF-Q): a cross-sectional study. *Health Sci Rep.* 2025;8(2):e70396.
49. Wilson M-MG, Thomas DR, Rubenstein LZ, Chibnall JT, Anderson S, Baxi A, et al. Appetite assessment: simple appetite questionnaire predicts weight loss in community-dwelling adults and nursing home residents. *Am J Clin Nutr.* 2005;82(5):1074–81.
50. Andreae C, Strömberg A, Sawatzky R, Årestedt K. Psychometric evaluation of two appetite questionnaires in patients with heart failure. *J Card Fail.* 2015;21(12):954–8.
51. Lovibond PF, Lovibond SH. Depression anxiety and stress scales. *Behav Res Ther.* 1995.
52. Buysse DJ, Reynolds IICF, Monk TH, Berman SR, Kupfer DJ. The Pittsburgh Sleep Quality Index: a new instrument for psychiatric practice and research. *Psychiatry Res.* 1989;28(2):193–213.
53. Węgrzynowska-Teodorczyk K, Mozdżanowska D, Josiak K, Siennicka A, Nowakowska K, Banasiak W, et al. Could the two-minute step test be an alternative to the six-minute walk test for patients with systolic heart failure? *Eur J Prev Cardiol.* 2016;23(12):1307–13.
54. Cocco G, Chu D. Weight reduction decreases NT-proBNP levels in obese coronary patients with chronic diastolic heart failure. *Arch Med Sci.* 2007;3(2):112–6.
55. Bui AL, Horwich TB, Fonarow GC. Epidemiology and risk profile of heart failure. *Nat Rev Cardiol.* 2011;8(1):30–41.
56. Stenholm S, Harris TB, Rantanen T, Visser M, Kritchevsky SB, Ferrucci L. Sarcopenic obesity: definition, cause and consequences. *Curr Opin Clin Nutr Metab Care.* 2008;11(6):693–700.
57. Vest AR, Schauer PR, Rodgers JE, Sanderson E, LaChute CL, Seltz J, et al. Obesity and weight loss strategies for patients with heart failure. *JACC: Heart Failure.* 2024;12(9):1509–27.
58. Zhao S, Dan L, Huang R, Shen Z, Huang D, Wu P, et al. Decoding the impact of gut microbiota on heart failure. *Genes Dis.* 2025:101592.
59. Shariff S, Huey AKS, Soni NP, Yahia A, Hammoud D, Nazir A, et al. Unlocking the gut-heart axis: exploring the role of gut microbiota in cardiovascular health and disease. *Ann Med Surg (Lond).* 2024;86(5):2752–8.
60. Khan MS, Khan F, Fonarow GC, Sreenivasan J, Greene SJ, Khan SU, et al. Dietary interventions and nutritional supplements for heart failure: a systematic appraisal and evidence map. *Eur J Heart Fail.* 2021;23(9):1468–76.
61. Sharifi-Rad J, Rodrigues CF, Sharopov F, Docea AO, Can Karaca A, Sharifi-Rad M, et al. Diet, lifestyle and cardiovascular diseases: linking pathophysiology to cardioprotective effects of natural bioactive compounds. *Int J Environ Res Public Health.* 2020;17(7):2326.
62. Ebong IA, Goff Jr DC, Rodriguez CJ, Chen H, Bertoni AG. Mechanisms of heart failure in obesity. *Obes Res Clin Pract.* 2014;8(6):e540–8.
63. Jamshed H, Beyl RA, Della Manna DL, Yang ES, Ravussin E, Peterson CM. Ten-hour time-restricted feeding improves 24-hour glucose levels and affects markers of the circadian clock, aging, and autophagy in humans. *Nutrients.* 2019;11(6):1234.
64. Wilkinson MJ, Manoogian EN, Zadourian A, Lo H, Fakhouri S, Shoghi A, et al. Ten-hour time-restricted eating reduces weight, blood pressure, and atherogenic lipids in patients with metabolic syndrome. *Cell Metab.* 2020;31(1):92–104.e5.
65. Moro T, Tinsley G, Bianco A, Marcolin G, Pacelli QF, Battaglia G, et al. Effects of eight weeks of time-restricted feeding (16/8) on basal metabolism, maximal strength, body composition, inflammation, and cardiovascular risk factors in resistance-trained males. *J Transl Med.* 2016;14:1–10.
66. Sutton EF, Beyl R, Early KS, Cefalu WT, Ravussin E, Peterson CM. Early time-restricted feeding improves insulin sensitivity, blood pressure, and oxidative stress even without weight loss in men with prediabetes. *Cell Metab.* 2018;27(6):1212–21.e3.
67. Anton SD, Lee SA, Donahoo WT, McLaren C, Manini T, Leeuwenburgh C, et al. The effects of time restricted feeding on overweight, older adults: a pilot study. *Nutrients.* 2019;11(7):1500.
68. Babalola OO, Akinnusi E, Ottu PO, Bridget K, Oyubu G, Ajiboye SA, et al. The impact of ultra-processed foods on cardiovascular diseases and cancer: epidemiological and mechanistic insights. *Asp Mol Med.* 2025:100072.
69. Tomova A, Bukovsky I, Rembert E, Yonas W, Alwarith J, Barnard ND, et al. The effects of vegetarian and vegan diets on gut microbiota. *Front Nutr.* 2019;6:447652.
70. Rowland I, Gibson G, Heinken A, Scott K, Swann J, Thiele I, et al. Gut microbiota functions: metabolism of nutrients and other food components. *Eur J Nutr.* 2018;57:1–24.
71. Lares-Michel M, Housni FE, Reyes-Castillo Z, Huertas JR, Aguilera-Cervantes VG, Michel-Nava RM. Sustainable-psycho-nutritional intervention programme for a sustainable diet (the 'NutriSOS' study) and its effects on eating behaviour, diet quality, nutritional status, physical activity, metabolic biomarkers, gut microbiota and water and carbon footprints in Mexican population: study protocol of an mHealth randomised controlled trial. *Br J Nutr.* 2023;130(10):1823–38.
72. Apalowo O, Adegoye G, Obuotor T. Exploring the functional roles of sustainable dietary patterns on gut microbiota composition and optimal health. *Bioact Compd Health Dis.* 2024;7(2):110–30.
73. Cahill LE, Hummel SL. Eating to prevent both heart and planetary failure: the EAT-lancet diet. *JACC Heart Fail.* 2024. pp. 1209–11.
74. Santiago-Torres M, Kratz M, Lampe JW, Tapsoba JDD, Breymeyer KL, Levy L, et al. Metabolic responses to a traditional Mexican diet compared with a commonly consumed US diet in women of Mexican descent: a randomized crossover feeding trial. *Am J Clin Nutr.* 2016;103(2):366–74.
75. Awoyemi A, Mayerhofer C, Felix AS, Hov JR, Moscovitch SD, Lappegård KT, et al. Rifaximin or *Saccharomyces boulardii* in heart failure with reduced ejection fraction: results from the randomized GutHeart trial. *EBioMedicine.* 2021;70.

Publisher's note

Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.